2024 SSMO Speed Round

SMO Team

SR 1: Find the sum of the distinct prime factors of $2024^2 - 1$.

SR 2: Gracie's students play with some toys. When 4 or 5 students are present, the toys can be equally distributed to everyone. However, when there are only 3 students, there is one toy left over after giving everyone the same number of toys. What is the least possible number of toys that Gracie could have?

SR 3: The polynomial $x^3 - 15x^2 + 4x + 4$ has distinct real roots r, s, and t. Find the value of

$$\left|(r^2 + s^2 + t^2)(rst)\right|.$$

SR 4: Sam wants to read the Harry Potter and Warriors books. There are 7 Harry Potter books that must be read in a specific order, and there are 6 Warriors books that also must be read in a specific order; however, he can read the two series at the same time. For example, he could read the first three Harry Potter books, then the first five Warriors books, then the remaining Harry Potter books, and finally the last Warriors book. In how many different orders can Sam read the books?

SR 5: Let $\triangle ABC$ and $\triangle ADC$ be right triangles, such that $\angle ABC = \angle ADC = 90°$. Given that $\angle ACB = 30°$ and $BC = 3\sqrt{3}$, find the maximum possible length of BD.

SR 6: There are 4 people and 4 houses. Each person independently and uniformly at random chooses a house to live in. The expected number of inhabited houses can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

SR 7: Let S denote the set of all ellipses centered at the origin with axes AB and CD, where $A = (-x, 0)$, $B = (x, 0)$, $C = (0, -y)$, and $D = (0, y)$ for integers x and y with $1 \leq x, y \leq 10$ and $2 \mid x + y$. Let T denote the set of all such ellipses with $2 \nmid x + y$. The positive difference between the sum of the areas of all ellipses in T and the sum of the areas of all the ellipses in S can be expressed as $m\pi$, where m is a positive integer. Find m.

SR 8: Bob has two coins; one is fair, and one lands on heads with a probability of $\frac{2}{3}$. Bob chooses one of the coins uniformly at random and flips it twice. Alice watches the two coin flips and guesses whether Bob flipped the fair or rigged coin. Given that Alice is a good mathematician and guesses the more likely option (guessing randomly when they are equally likely), the probability she guesses right can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

SR 9: Let a, b, c, and d be positive integers such that $abcd = a + b + c + d$. Find the maximum possible value of a.

SR 10: Let $a_1, a_2, \ldots, a_{14}$ be the roots of $(x^7 - x^3 + 2)^2 = 0$, counted with multiplicity. Find the value of $\prod_{i=1}^{14}(a_i^7 + 1)$.